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Using virtual reality to examine perceptions of coherence, complexity, and time with vegetation layout

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ABSTRACT

Designers influence people's perception of time and experience of place through the built environment, yet the consequences of particular design decisions remain mostly elusive. This study explores the use of virtual reality to investigate the relationship between site users' perceptions of time and landscape aesthetics, specifically the coherence and complexity of an outdoor planting design. An experimental research design included four groups of study participants moving through a modelled campus path with different cases of vegetation layouts. Then, the participants took part in structured focus group interviews ($n=16$). Analysis included in-depth thematic qualitative coding of the textual information as well as computational text assessments, and findings were corroborated with existing literature to consider the soundness and potential benefits of this methodological approach. This study suggests how virtual reality can be used to isolate the design variable of vegetation to better understand the outcomes of planting design decisions.

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Introduction

The presence of certain elements within a landscape can alter a user's perception of time moving through a designed space (Abdelrahman et al., 2018). How people perceive time can dictate opinions formulated about a landscape and relate to human enjoyment of a space depending on how time passes, for example, quickly through contentment or slowly through leisure (Spooner, 2011). Studies have also illustrated users' perceptions while they move along paths as being defined by differing facets including patterns, lack of disturbances, and unified elements (Kaplan et al., 1998; Spooner, 2011; Yan et al., 2023). The layout of site elements contributes to the perceived coherence and complexity of the landscape, as the presence of patterns, rhythm, and spatial complexity affect the subconscious experience of site users. While these impressions often happen without notice, designers may make intentional design decisions surrounding landscape aesthetic principles with specific outcomes in mind.

Research on landscape aesthetics has been performed for decades, yet many approaches have posed significant limitations. For example, site observation is often costly and time-consuming, and photo elicitation can be influenced by many factors such as the angle of

view (Sevenant & Antrop, 2011). Studies have used 3-dimensional models of site elements for perceptions (Kuper, 2017), yet scenes have still been shown in a static, 2-dimensional format. While emerging methods such as the assessment of social media data show promise in understanding human perspectives of the aspects of a place (Fernandez et al., 2022; Song et al., 2020, 2023), they still do not provide first-hand information on people's perspectives of site design considerations. Therefore, new methods should be examined which can more specifically consider the influence of site design variables.

This study explores the use of virtual reality (VR) in landscape aesthetics research by implementing the controlled manipulation of a design variable. More specifically, VR was used to investigate how changes in vegetation along a campus path influence people's perception, focusing on the relationship between levels of coherence and complexity in planting layouts and site users' sense of time. This study expands on prior research, including: Kuper's (2017) examination of planting layout in relation to landscape preference, coherence, and complexity using photo elicitation techniques through site modelling graphics; Bosselman's (1998) analysis of environmental perceptions through movement along a path, captured via sequential landscape snapshots; and Isaacs' (2001) investigation into how the physical characteristics of streets influence individual perceptions of the passing of time. Additionally, vegetation has been widely studied for its aesthetic and environmental benefits (Du et al., 2016; Zhuang et al., 2021) but new approaches could expand research and provide different types of information.

Findings from this exploratory study were substantiated by results from the existing literature previously collected by non-VR means, suggesting that VR may provide a useful avenue for considering the impact of design decisions such as planting schemes. Overarchingly, this study shares a novel approach for investigating aspects of landscape aesthetics through virtual reality that may allow a deeper understanding of the potential impact of particular design variables.

Virtual reality as a measurement tool for landscape aesthetics

VR has the potential to create high-quality environmental simulations and human-space interactions in a virtual environment, providing researchers the ability to study landscape aesthetics and the people-environment relationships in a more detailed and determinate way. It enables participants to feel as though they are actually in the space that they are engaging with digitally (Ahn et al., 2022). In its early conception, VR was predicted to become a measurement tool in aesthetics research (Daniel, 2001), yet this opportunity has not been fully realised.

VR has been used to address a range of human-environment interactions, including human and environmental health issues (Vercelloni et al., 2018), sustainable community design (Maffei et al., 2016), and applications for design decision-making (George et al., 2022). Regarding landscape aesthetics assessment, VR has been used to evaluate the aesthetic experience of architectural spaces such as rounded, curvilinear and sharp, and rectilinear elements (Carreiro et al., 2017). Another study developed three scenarios of the same virtual park with varying atmospheres such as shadows, lights, and textures, and evaluated the emotional variances experienced by participants during immersive encounters (Riva et al., 2007). VR has been used to understand the relationship between perceived safety and a designed versus an undesigned alleyway, modifying the site elements of seating, lighting, planting, and shade structures (Martinez et al., 2024). Finally, the technology has also been used to assess the validity of on-site surveys compared to VR for assessing aesthetic preferences, where the results supported that VR can be considered as a reliable tool for assessing aesthetic preferences as on-site surveys (Shi et al., 2020). However, to date and to the authors' knowledge, VR studies have not isolated the design variable of vegetation in the landscape related to planting design layout to understand its relationship with perceived site experiences.

Defining coherence and complexity

Elements in the landscape can determine the rhythm of a place and impact the perceived quality and character of a landscape. As a part of landscape aesthetic theory, coherence can be defined as the relative amount of unity in a scene and complexity as diversity and richness within a landscape which affects landscape character and visual preference (Ode et al., 2008). Designers can place these elements intentionally to craft the level of coherence and complexity and define the experience of site users. In doing so, these elements can be manipulated to create patterns (Bosselmann, 1998). Coherence and complexity exist broadly throughout the world, and when applied more specifically to an area of study such as landscape elements, these elements dictate different environments (Figure 1).

Designed coherence exists in the landscape through distinct regions and unifying textures and themes, whereas designed complexity evinces itself through variety and richness throughout the landscape (Kaplan et al., 1998). For this study, coherence is defined as the level of order and organisation, and complexity is defined as the presence or lack of variety regarding plant species. These definitions can describe types of planting arrangements and quantities, such as the low coherence scattershot placement that possesses no perceivable pattern and the high coherence formal placement that utilises the patterned placement of plant species (Kuper, 2017). Complexity can then be determined by the number of plant species that afford it high or low variety.

Coherence and complexity both contribute to a user's comprehension of the environment as coherence enables understanding to occur through grouping and categorisation, defining the landscape through order or chaos of plant groupings. Complexity enables exploration through variety and levels within ordered or unordered groupings (Kaplan et al., 1998). Coherence and complexity can dictate the degree of order people experience within a landscape which prompts the question of how varying degrees of order influence a site user's perception of an environment, and if coherence and complexity establish rhythms and patterns that affect the rate at which users experience and perceive time.

Complexity's effects on time and perceived duration have been analysed and translated into landscape experiences. For example, adding more complexity to an environment has the potential to lengthen the perceived passage of time by enhancing visual depth and layering elements to create hierarchy and a sense of suspense (Isaacs, 2001). Extreme levels of complexity, whether high or low, are associated with a longer perceived duration of time with a moderate level causing the duration of time to lessen (Shakibamanesh & Ghorbanian, 2017). With this observed precedent, an environment with a moderate level of complexity would shorten a user's walk and any variation in complexity, either greater or less than, would lengthen the walk. However, the relationship between user perceptions and varying levels of

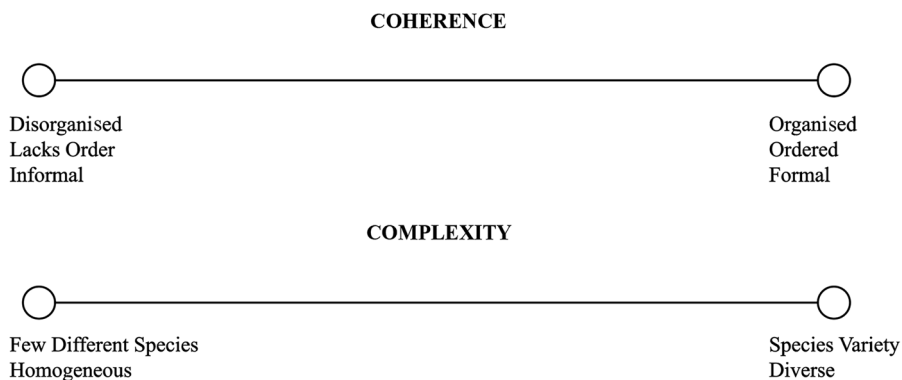


Figure 1. Characterisation of coherence and complexity.

complexity have not been tested in an immersive environment, a limitation that is noted in the literature.

The phenomenon of experiencing time

Time is a complex phenomenon that can be difficult to measure (Isaacs, 2001). Studies touch on aspects of time relative to landscape experiences (Yang et al., 2007), yet elaboration beyond acknowledgment of the phenomenon remains vague (Fraisie, 1984). Time is incorporated into people's experiences whether perceptual, intellectual, or emotional (Jones & Meagher, 2015). The experience of time is different from taste, smell, and other senses, yet humans 'feel' it from a personal vantage point (Ornstein, 1969). Its relative nature grants it an elusive quality and makes its perception difficult to truly understand and measure (Jones & Meagher, 2015).

To experience time is to perceive change and events in succession (Isaacs, 2001). The repetition of these events and elements creates patterns that enable the user to more readily perceive the passage of time, creating a means for designers to decide how a user will experience the passage of time through a particular space (Bosselmann, 1998). In this context, user experience specifically regarding time can be altered to achieve design goals for how a space can be experienced as a person interacts with an environment through movement. Designers may be able to create goals to utilise time and its related variables throughout the landscape in terms of speed, intentionally creating the feeling of longer or shorter experiences within a designed landscape (Isaacs, 2001). For example, when walking, low information trips with less features are theorised to seem slower than high information trips (Rapoport, 1977). Manipulating the physical environment to adjust the perception of time initiates many interesting design possibilities from an experiential perspective (Spooner, 2011). Yet, little specific direction has been given on how to achieve design goals related to the passing of time.

Considering motion through landscapes

Movement through a landscape is often the way that an experience takes place (Jones & Meagher, 2015), and different means provide varying levels of interaction with the environment (Rapoport, 1977). For example, a stroll on foot affords a person a high level of immersion with the surrounding place. Circulation through a landscape goes beyond the routes people take, intertwining with a person's experience of a landscape where certain paths can connect to emotions (Mosleh, 2019) or convey memories, stories, or feelings. Elements along paths for any means of transportation combine with motion to dictate certain realisations. Furthermore, vision provides a sense of motion, enabling the user in motion to understand displacement through the apparent movement of nearby objects that they know to be fixed to be the result of their own advancement (Appleyard et al., 1965). While people interact with the environments around them through motion constantly, they tend to neglect this connection. Bosselmann (1998) was particularly interested in studying experiences moving through a site, though he acknowledged the need for a 360° visual experience that one would enjoy during a physical walk along the path.

Methods

This study took a sequential mixed-methods approach with an exploratory experimental research design. Participation in this study was limited to students and faculty at the university who were 18 years of age or older. The study was approved by the Institutional Review Board at the university where the study took place, and the authors report there are no competing interests to declare. The researchers adopted an interpretivist approach, focusing on uncovering the

meanings participants assigned to their experiences. The researchers acknowledge that their findings are inherently subjective, influenced by their backgrounds as well as the cultural and social context of a university setting in the southeastern United States. Rather than seeking objective or universal truths, the study aimed to capture the depth and nuance of participants' experiences within a virtual environment, and to test the methodology of exploring the isolated variable of vegetation in VR.

3-Dimensional model and virtual experience

A 3-Dimensional (3D) model was generated to produce four identical campus path scenarios with differing levels of coherence and complexity. Four distinct environments were created based on previously tested metrics for studying the effects of manipulated amounts of designed coherence and complexity in landscapes (Kuper, 2017) (Figure 2). A high level of coherence was modelled using the discernible presence of a few distinct regions and increased quantities of repeating themes and unifying textures, while a low level of coherence included little perceivable pattern of vegetation within the landscape. High levels of complexity were modelled as environments with many different visual components to consider and low levels of complexity were found in large, open expanses with lower levels of variety. The environments included paths of identical length surrounded by contrastingly coherent and complex designs in the surrounding landscapes.

Each of the four models was rendered out as a two-minute-long 360° video using Twinmotion visualisation software. The video exports were then viewed in VR using HTC Vive Cosmos headsets and consoles through the Viveport Video application by four groups of study participants. Each group had four study participants for a total of sixteen participants (n=16), and the groups took the experiment separately. Movement along the path was replicated identically for each scenario with participants asked to watch the VR video twice to reduce the novelty effect of VR. Participants viewed the environment from the fixed vantage point of moving along the path. To achieve the effect of a campus path, the model was built similar to an existing campus layout at the researchers' higher education institution (Figure 3). Plantings framed the paths at the same points along each path, where only plant variety

	Low Coherence	High Coherence
Low Complexity	Environment 1 <i>Low Coherence</i> <i>Low Complexity</i>	Environment 2 <i>High Coherence</i> <i>Low Complexity</i>
High Complexity	Environment 3 <i>Low Coherence</i> <i>High Complexity</i>	Environment 4 <i>High Coherence</i> <i>High Complexity</i>

Figure 2. Combination of coherence and complexity levels within modelled VR environments.

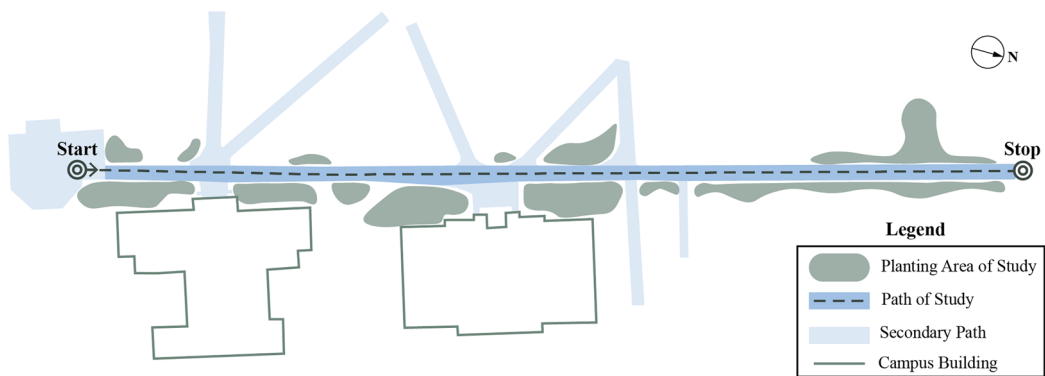


Figure 3. Map depicting plant massing layout zones along campus walking route, with walking route approximately 200 metres long.

and organisation or quantity were altered to generate distinct levels of coherence and complexity.

Prior to the VR experience, participants were given the definitions for coherence and complexity and asked to pay particular attention to the vegetation along the path. The number and organisation of plants used in each model was based on Kuper's (2017) study, which differentiates levels of complexity with set numbers of plant species and levels of coherence with different organisation patterns (Figure 4). Low coherence scattershot placement possesses no perceivable pattern, while high coherence formal placement utilises patterned placement of plant species. The planting bed locations mimicked the visual planting bed imagery in the precedence study (Kuper, 2017), as well as existing site conditions. The final layouts were reviewed by a professional landscape architect and campus planner for quality control of the planting design in relation to plant layout and the representation of plants in the virtual models.

Focus group discussions

After interacting with one of the four environments in VR, each group engaged in an audio-recorded structured focus group discussion surrounding their thoughts and experiences about the VR environment they went through (low coherence/low complexity; high coherence/low complexity; low coherence/high complexity; or high coherence/high complexity). Focus group questions were formulated deductively based on pre-determined concepts to elicit responses regarding participants' perceptions of time, coherence, and complexity, and the interrelations between these variables. More specifically, the focus group questions were modified from Kaplan et al.'s (1998) study (Table 1).

Data analysis was performed through transcriptions of conversations and thematic qualitative coding of the textual information using NVivo software. Thematic coding of qualitative information includes identifying text that exemplifies a relevant theme and linking portions of text through codes (Gibbs, 2007). Themes for analysis were determined based upon categories derived from research questions and narrowed for relevance (Vaughn & Turner, 2016). This paper adheres to the Standards for Reporting Qualitative Research (O'Brien et al., 2014).

To gather rich qualitative data, one primary researcher actively engaged with participants, facilitating a deeper understanding of participants' perspectives which also allowed for the researcher's own interpretations and reflections to be thoughtfully and transparently integrated into the analysis. This researcher performed the first round of coding, considering the general thoughts and emotions that were perceived in person through body cues and vocal tone. The focus group interviews were assessed within 2 weeks of their completion, and the researcher

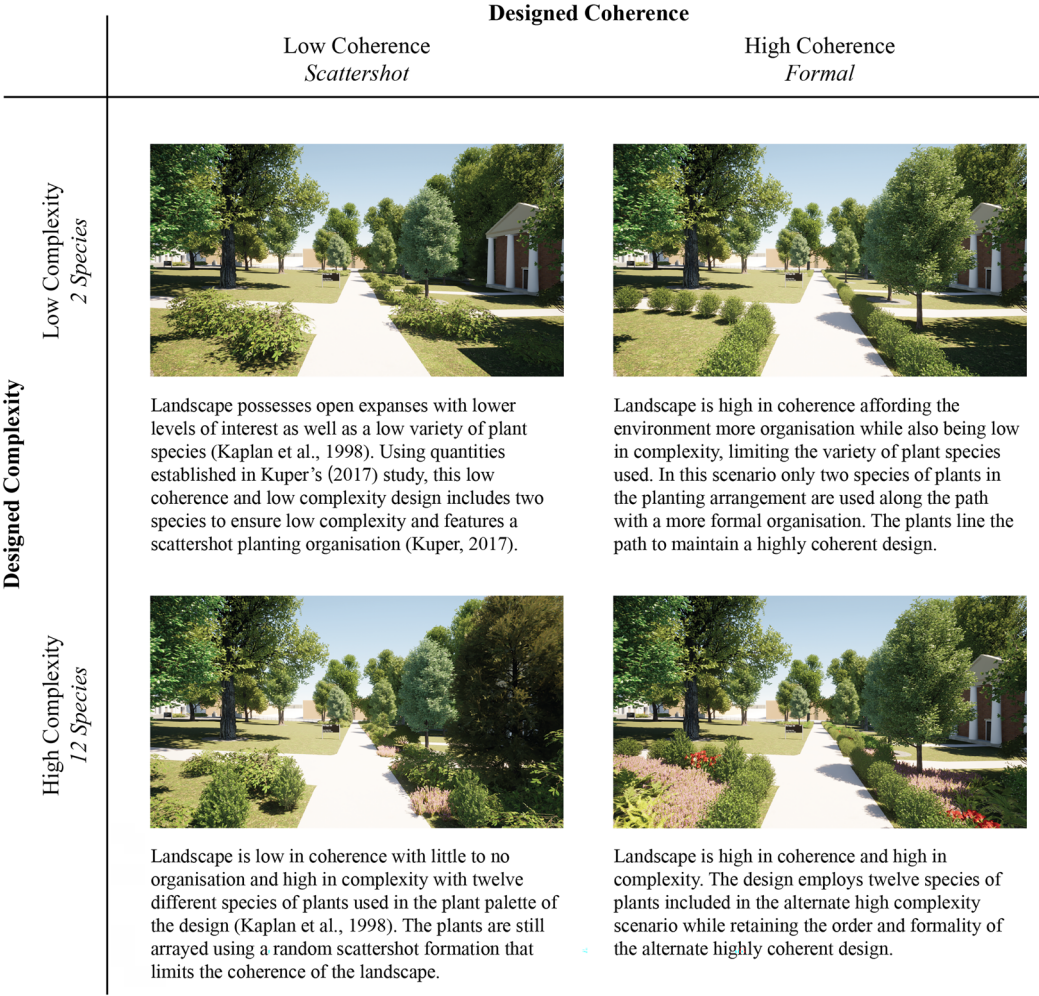


Figure 4. Chart depicting categorisation of VR experiences, modelled after levels of coherence and complexity used in Kuper (2017).

Table 1. Focus group questions asked to qualitatively measure different facets of the study.

Focus Group Discussion Protocol	
Moderated Questions	Measure
How much is going on in this scene?	coherence
How much is there to look at?	coherence and complexity
How much does the scene contain many elements of different kinds?	complexity
How would you rate this environment in terms of coherence on a scale of 1–5?	coherence
How would you rate this environment in terms of complexity on a scale of 1–5?	complexity
What are three adjectives you feel embody this environment?	open qualitative question
At what pace do you feel that time passed?	perception of time

had access to the audio recordings of the focus groups as well as notes from the interviews to help with understanding the context of the transcripts. To increase the reliability of findings, another researcher provided an in-depth intercoder agreement check of the first investigator's output compared to the data presented (Huberman, 2014). Because qualitative findings involve interpretation and contextualisation, evidence such as examples, quotes, and text excerpts were noted to support the researchers' understandings (O'Brien et al., 2014).

As a supplementary form of analysis, a computational text assessment was applied to the transcripts through word frequency counts, a technique that enables computers to count word occurrence in a text set (Hai-Jew, 2022). For this study, words with similar meanings were grouped together as specialisations. The 30 most frequently used words/specialisations were taken from focus group transcripts for each coherence and complexity case using NVivo software, and the top 10 were used for analysis. Stop words such as subject pronouns, non-action verbs, and adverbs including 'like', 'actually', and 'during' were considered irrelevant and removed to not alter the hierarchy within the qualitative assessment of word frequencies.

Results

VR video experiences were conducted for the four participant groups, followed by focus group discussions. All groups were mixed with both females and males between the ages of 18 and 35, and all participants were university students from undergraduate and graduate levels. Females made up 62.5% of total participants, and males 37.5%. The study took approximately 30–40 min for each group which included signing waivers and a pre-study briefing, running through the 2-minute VR 360° video twice on a loop, and then in-depth focus group discussions.

Exploration of vegetation coherence and complexity combinations

Low coherence/low complexity

In the scenario with both low coherence and low complexity, the path experience was considered sparse and under-stimulating. All participants described the frequent repetition of the plant species, with various comments reflecting this outlook such as, 'I saw the same bush all over the place... it didn't feel like there was too much going on because I just felt like a lot of the planting was the same'. One participant explained that due to a lack of stimulation, they were hoping for a change but did not experience one. All participants noticed and described details beyond the adjacent plantings, such as trees in the distance, architectural elements, and the spaces between buildings, despite the suggestion to focus on plantings immediately along the path. The word frequency of the associated focus group discussion also reveals a concentration on 'planting', 'trees', and 'different' (Table 2), which when combined with other qualitative assessments, reveals the desire for more differences in these areas.

Most participants agreed that the experience overall felt 'on the slow side'. The second loop of the video was generally perceived to be slower than the first, most likely due to the lack of visual stimulation as the plantings were dominated by a single type of shrub. Participants also described that they felt isolated, as they wanted to move beyond the path and take connecting paths that veered off. Participants mostly contributed this to the plantings along the path which did not offer interest to keep them engaged and satisfied as they moved along. The scene was described as a pass-through space: 'It was transitional... it felt like if I were in that spot in real life it would be like I'm walking through the space and there's not really somewhere that I need to stop and stay'. Time was perceived to pass more slowly due to this absence of interest and participants desired to spend less time in the 'transitional' space.

Despite the discussion surrounding a lack of stimulation and slowness, the researcher generally perceived the tone of remarks and descriptions of the space as somewhat positive. When asked for three adjectives that embodied the environment, participants focused on qualities of the experience using the words 'calm', 'quiet', and 'relaxing'. This contrasts with the tidiness, sterility, and formality associated with the high coherence scenarios discussed later in this section. One participant described the path similar to 'spaces to get away from your senses... this is not in a bad way but [in an] under-stimulating, relaxing kind of way'. Based on the researcher's engagement with the participants, it was interpreted that their experiences were neither negative nor especially stimulating.

Table 2. Top 10 word frequency with specialisations from focus group transcripts for each model case.

Low Coherence/Low Complexity				High Coherence/Low Complexity			
Word	Ct.	Wt. %	Similar Words	Word	Ct.	Wt. %	Similar Words
Planting	15	5.38	Plant, plantings, plants	Ordinary	10	5.45	Average, mundane
Trees	12	4.3	Corner, tree	Boring	7	3.94	Slow
Different	8	2.87	Difference	Different	6	3.64	Difference, otherwise
Space	8	2.87	Distance, spaces	Plantings	6	3.64	Plant, planting
Beds	7	2.51	n/a	Trees	6	3.64	Tree
Bush	7	2.51	Bushes	Shrubs	5	3.03	Shrub
Slow	7	2.51	Dense, slowly	Directed	5	2.73	Direct, guided, organised
Buildings	7	2.33	Building, form	Time	4	2.42	Times
Noticed	7	2.33	Note, notice, noticing, obviously	Building	4	2.12	Buildings, makes
Coherent	6	2.15	Coherence, coherency, cohesive	Catch	3	1.82	view, watching

Low Coherence/High Complexity				High Coherence/High Complexity			
Word	Ct.	Wt. %	Similar Words	Word	Ct.	Wt. %	Similar Words
Diverse	6	4.8	Diversity	Plants	9	4.59	Planting, plantings, setting
Repetitive	6	4.8	Repeating, repetition	Repetitive	7	3.57	Repeating, repetition
End	5	4.0	Completely, ended	Slow	7	3.57	Dense, slowed, slowing
Plant	5	4.0	Plantings, plants	Time	5	2.55	Times
Different	3	2.4	n/a	Walking	6	2.55	Passed, walk
Good	3	2.4	full	Coherent	5	2.3	Consistent, orderly
Now	3	2.4	n/a	End	4	2.04	Stop, stopped
Organised	3	2.4	Organisation	Faster	4	2.04	Quicker
Satisfying	3	2.4	n/a	Life	4	2.04	Animals
Space	3	2.4	Distance	Moving	4	2.04	n/a

Legend

Average Weight Percentage (%)

5.00+

4.00–4.99

3.00–3.99

2.00–2.99

High coherence/low complexity

In the scenario with high coherence and low complexity, the scene was overwhelmingly perceived by participants to be uninteresting and ordinary. The researcher observed that participants' experiences lacked the sense of satisfaction and calm that the low-coherence, low-complexity scenario provided, instead being notably strong and negative. For instance, when asked to provide three adjectives that embodied the environment, several participants repeatedly and emphatically stated the word 'boring'. All but one participant strongly agreed with this assessment, while the exception preferred the term 'mundane', explaining that although they did not feel bored, the scene lacked any 'wow factor'.

Related to time, the experience was generally described as slow and redundant. One participant summarised this sentiment by stating, 'I thought I was in there for a while' and 'that was the longest two minutes I've ever felt'. Participants attributed this perception to the repetitive presence of a singular, uninteresting shrub, which led them to quickly lose interest in the immediate scene and instead direct their attention towards alternative path options and the more visually engaging trees in the distant quad. In fact, one participant described the trees as more interesting due to their scattershot placement: '...the trees were just scattered out in the middle of lawn and they give a bit more character whereas the shrubs were... just kind of like somebody clicked the offset button from the sidewalk'. Additionally, participants noted that the second viewing of the video felt noticeably slower than the first, reinforcing their sense of monotony. They also agreed that, in a real-life experience, they would have moved through the space more quickly, a reaction similar

to that of participants in the alternate low complexity scenario. One participant stated, 'I definitely feel like I would walk through that space faster normally'; which was echoed by the rest of the group.

Finally, participants perceived the environment as structured, particularly through the role of planting arrangements and the overall sense of coherence and organisation. Participants noted that the shrubs along the walkway played a defining role in directing movement, with one stating, 'the shrubs really lined and really dictated where you were going', while another described that 'I thought [the experience] direct... it guided [me] through the path'. Descriptive adjectives such as 'formal' and 'directed' further emphasise that the space was structured rather than chaotic. The word frequency assessment overarchingly supported the results, with the top words being 'ordinary' and 'boring' (Table 2).

Low coherence/high complexity

The researcher holding discussions observed that participants generally responded positively to the low coherence, high complexity scenario, reflected in their responses and optimistic tone. They implied that the plantings contributed to a sense of comfort and engagement by offering a balanced level of variety. In particular, safety emerged as a descriptor of the experience, with one participant stating, 'safety...I just like a lot of plants put together more', and another adding, 'it just feels like a safer environment'. The vegetation was described as 'pleasant, not overbearing', also potentially contributing to comfort and a sense of security, with more diverse plantings in a looser style when compared with the other scenarios.

Participants explicitly described the virtual experience as more favourable than the present real-life conditions based on its ample vegetation, describing it as 'easier to read' and that 'I definitely feel like I would like this more than what we have now'. They characterised the space as a 'satisfying' path that was 'full' and 'a space that [one] could walk through nicely', indicating greater satisfaction and enjoyment. This depiction suggests preference of a leisurely stroll through a soft and 'less intense' space instead of a structured walk. When asked broadly about what there was to see, participants primarily focused on the vegetation including the 'interesting blooms and colour'. Adjectives such as 'continuous' and 'unified' were interpreted as more positive descriptions of perceived lower levels of order and repetition within plantings in a low coherence scenario based on vocal tone and context. This contrasts with the 'structural' and 'architectural' consistency of the high coherence, high complexity scenario, as discussed later in this section.

After watching the video twice, the second video was described as feeling faster than the first which was also found in the other high complexity scenario. Participants credited this change in speed to a familiarity with the plantings present in the VR video and knowing how long the video would go. The second time, participants described that they focused on the known areas of plantings and felt as though they had enough visual stimuli to engage them throughout their walk. The word frequency assessment reinforced participants' discussion surrounding the diversity of the VR environment compared to the real-life scenario, with the most frequent mention being 'diverse'.

High coherence/high complexity

The researcher conducting focus group discussions observed that participants generally perceived the path scenario with high vegetation coherence and complexity as neutral to somewhat negative, based on their comments and attitudes towards the experience. Participants confirmed that sections of vegetation possessed a high level of complexity, stating that 'there was [a] decent amount of variety' and that 'individual pockets [of vegetation] had complexity'. However, their discussion suggested that the high levels of coherence and subsequent presence of repetition overpowered the complexity, with the order masking the plant variety. One participant described that the 'purple flowers felt... almost like a landmark...there was a lot of repetition of the same' while another shared that 'it felt like I was only looking at them again and again to see if there

were differences but there weren't...it felt a little repetitive, especially the evergreen hedges'. To further substantiate this finding, the word frequency analysis indicated that the term 'repetitive' was one of the second most frequently mentioned words, following 'plants' (Table 2).

The planting beds captured participants' interest to count and identify species and notice changes in colour and form, which contrasted with the low complexity scenarios that enabled users to quickly recognise the repetition of a single species and shift their interest to distant stimuli. Yet, the green hedges associated with high coherence were generally disliked, with one participant describing them as 'very stagnant... the green meatballs' and another nodding and repeating this descriptor with a look of distaste. Participants generally agreed that they 'have a preference for really wild looking [vegetation]' which would lean towards a less coherent vegetation layout. When asked for adjectives that embody the environment, participants provided words that the interviewing researcher perceived were being presented in a negative connotation including 'structural', 'tidy', 'sterile', 'repetitive', and 'impersonal', with milder remarks targeting the scene as not feeling 'terribly complex'. The participants expanded on these thoughts with explanations such as, 'neat in like the tidy sense, not like the cool version of that' and 'I almost want to say... there wasn't a ton of personality to it...the adjective would be...impersonal maybe'.

Participants generally felt that time passed 'quicker than you would experience in real life'. One participant stated that 'I would have stopped and looked at things a little in more detail... it felt like as soon as I looked down to look at something I was already passed it'. Participants wanted to observe the complexity within the scene, indicating higher levels of interest and a desire to enjoy the space at a slower pace. The word 'slow' was among the top three mentioned in the word frequency assessment, corroborating analysis surrounding the high coherence/high complexity environment, where 'slow' in context can be interpreted as a desire to see the landscape more slowly in the setting (Table 2).

Using virtual reality to isolate and evaluate vegetation in design

The VR experience allowed participants to analyse planting design layouts in a controlled and immersive environment. Many participants noted the high level of realism in the VR setting, which closely resembled familiar landscapes. One participant stated, 'I thought that it looks very much like campus... it felt like I was walking through the quad', while another commented on the accuracy of built elements, saying, 'It's crazy how real the buildings look... they look exactly like how they look [in real life]'. This realism enabled users to concentrate on the intended design elements without the distractions of an unfamiliar setting, allowing vegetation to emerge predominantly as the variable of interest when participants engaged with the planting design. Additionally, participants recognised specific trees within the virtual landscape, reinforcing the accuracy of the representation.

Repeated exposure to the VR environment as part of the research design improved participants' engagement with the space and helped them adjust to the technology. With some initially focused on navigation, users expressed that they were able to become more attuned to design elements. As one participant noted, 'The first [round] I was adjusting... I would turn and the plant would be there'. Another described how their focus evolved: 'I focused on like the path and then... realised the second go around that I could look around more'. These responses suggest that increasing familiarity with VR during the experiment enhanced participants' ability to concentrate on the environment as intended.

Based on focus group discussions, participants' perceptions of movement and spatial relationships evolved through the VR experience, reinforcing its potential as a tool to understand design variables. One participant stated, 'It was interesting to see the difference between the foreground and the background and the plantings within that space', demonstrating an awareness of spatial organisation and vegetation distribution. Another noted, 'I think it provided a good understanding of how it could be in that space', suggesting that VR offered a clear

representation of the designed environment. Participant responses suggest that VR effectively emphasised the planting design layouts, enabling participants to engage with vegetation composition and spatial relationships in an immersive, realistic, and analytical manner.

Participants largely adhered to the instructions by focusing their discussions on vegetation, even though they had the freedom to comment on other elements of the scene. While some participants noted other aspects, such as buildings or pathways, their observations often circled back to the plantings, indicating that vegetation was the dominant design variable shaping their experience. A few participants even mentioned that they attempted to focus on non-vegetation elements but found them less engaging or less varied, as reflected in comments like, ‘I wanted there to be more depth’, and ‘I wasn’t so interested in seeing the same buildings’. Others thoroughly described the layouts of plant materials, particularly shrubs, and how this influenced their perception of the space. This suggests that the VR environment successfully isolated vegetation as the key design variable, as participants naturally oriented their discussions around it, reinforcing the intended focus of this research effort.

Discussion and conclusion

For this study, VR was explored as a tool to isolate a particular site design variable, vegetation, to better understand its relationship with site users’ experiences and perceptions of time. The coherence and complexity of site vegetation were found to have unique relationships with different components of user experience, and patterns arose that serve as suggestions for researchers and designers of outdoor spaces. This section corroborates findings with the existing literature, including relationships between coherence and complexity and perception of time. Furthermore, it indicates design directions based on these substantiated findings (see [Figures 5–7](#)).

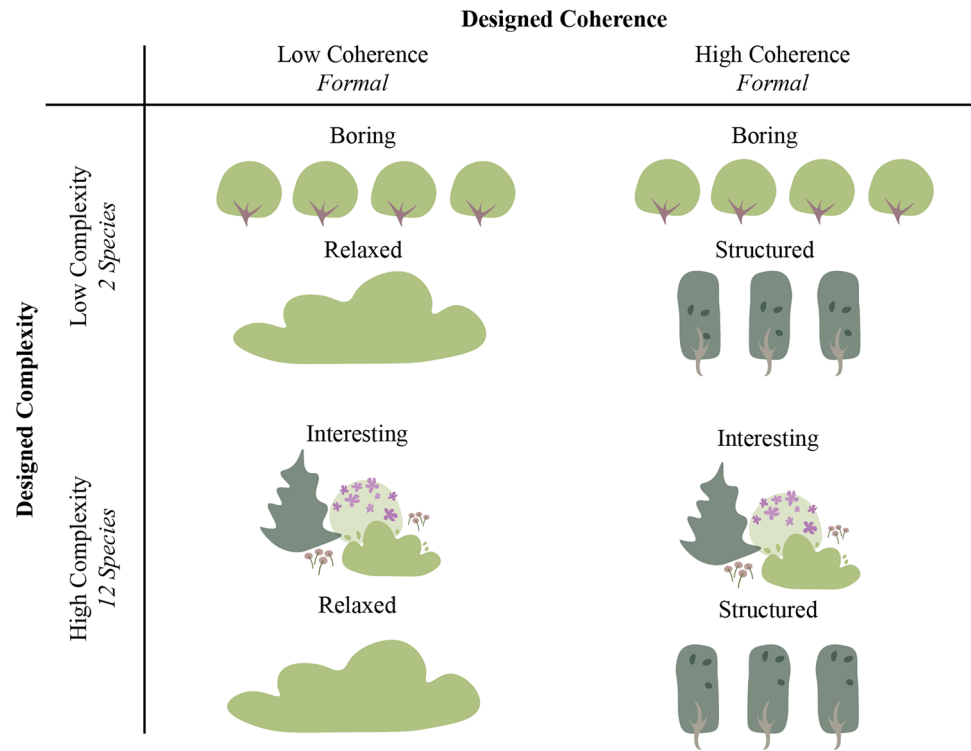


Figure 5. Diagrammed relationship between designed levels of coherence and complexity and characteristics of planting.

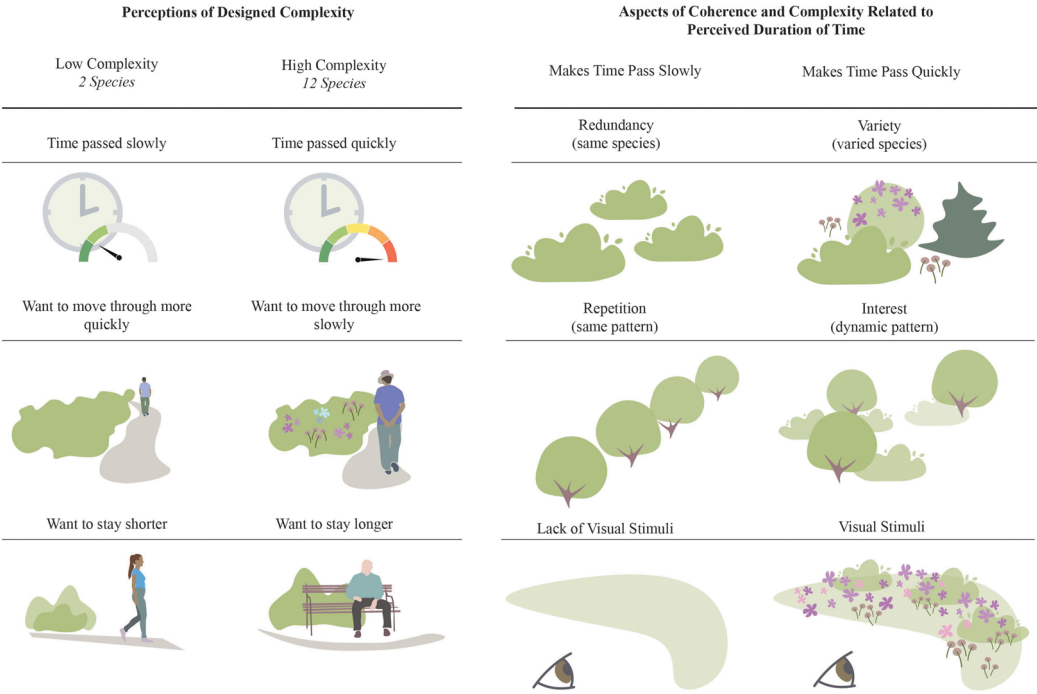


Figure 6. Diagrammed relationships between designed complexity and time, and relationships between aspects of coherence and complexity compared to perceived duration of time.

Virtual reality in landscape aesthetics research

Landscape aesthetics principles often relate to how humans perceive site layouts and elements, which is critical for aligning design intent with site use (Song et al., 2020). In this study, VR was used to isolate a specific design variable, vegetation, while factors like the surrounding environment and movement speed remained constant. Unlike 2D methods such as photo elicitation, VR offers a more immersive experience, better replicating findings from on-site observations and interviews (McLaughlan, 2019). For this research, VR was chosen to see how it might corroborate with existing studies on coherence, complexity, and time in a varying landscape vegetation layout.

The use of VR and modelling programs in this study provided several practical benefits. Once the base 3D model was created, design variables could be easily adjusted to develop four scenarios exploring different planting schemes based on coherence and complexity, with vegetation changes informed by Kuper (2017). VR enabled testing these distinct vegetation layouts in an immersive pilot study and quickly responding to feedback, tasks difficult to achieve in real-world settings. The flexibility of VR allowed the researchers to test these multiple scenarios while maintaining an experience more closely mimicking real settings than other methods. VR has been predicted to become a preferred method over on-site surveys and other methods due to its speed, cost-effectiveness, and efficiency (Watson & Petrie, 2010), and these benefits were realised in this exploratory effort.

Motion sickness in VR immersion has been noted as a concern in landscape research (Liu et al., 2023). However, in this study, all participants completed the research without issues, and no VR-related problems were raised during focus group discussions. Despite being informed of their option to withdraw, no participants chose to leave the study. The researchers observed that students were eager to use the equipment and engaged actively in follow-up discussions. The use of 360-degree VR videos exported from Twinmotion may have been a

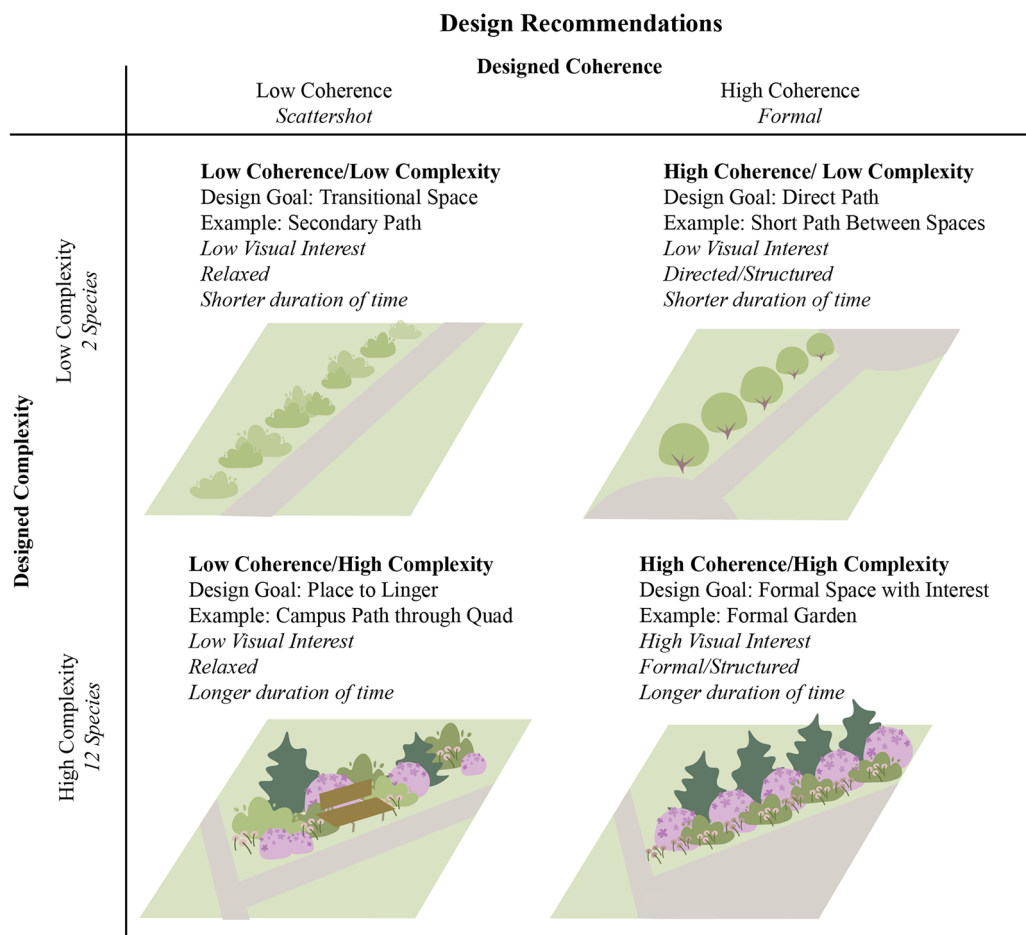


Figure 7. Design directions based on relationships found through analysis of qualitative discussion and corroboration with literature.

more reliable approach than direct immersion in more complex VR programs, as it employed a VR video player that was not prone to technical issues during data collection. The method allowed participants to experience the site twice automatically, which may have helped them form a more comprehensive understanding of their virtual surroundings. This immersive VR video approach could serve as a valuable model for future research, especially in studies requiring consistency in time or spatial boundaries, such as those with a set path or viewing duration.

Although participants in this study were assigned to a single environment with one changing variable, other studies using VR to display real environment imagery have incorporated multiple scenes, such as different seasons (Xiang et al., 2021) or varying landscapes (Shi et al., 2020) and gathered reviewer feedback. As VR technology becomes increasingly accessible (George et al., 2022), this study demonstrates the potential for using this approach to gain a deeper understanding of design variables. For example, with greater power over control variables, VR could be employed to explore causal relationships between specific design elements and user outcomes. Moreover, some VR studies have integrated the collection of physiological data when viewing virtual landscapes (Seiz et al., 2023), opening up possibilities for combining physiological and perceptual data to better understand the impact of design elements on site users.

Suggestions from the study

Coherence was shown to contribute to the familiarity of a space while complexity contributes to the perceived levels of interest. Higher levels of coherence through formal planting layout and organisation were associated with an experience that was structured, predictable, and repetitive while lower levels of coherence were related to a relaxing and satisfying scene in a more familiar environment. Higher levels of complexity generated higher levels of visual stimuli that offered variety and interest adjacent to the path. Lower complexity was contrastingly perceived as underwhelming and boring, prompting visual and sensory exploration beyond the plantings along the path (Figure 5). In the focus group with high coherence and complexity, participants described the experience as repetitive and impersonal, which did not emerge in the low coherence high complexity group. This may suggest that coherence is more powerful in shaping perception than complexity since the order and layout of plantings appeared to more strongly influence participants' perceptions than the variety of plant types in focus group discussions for these two cases.

These impacts on experience were shown to relate to the perception of time with travel feeling as though it passed generally faster in high complexity environments and slower in low complexity environments. Scenes with high complexity were also identified as places to spend additional time in or move through more slowly, while scenes with low complexity were identified as spaces to spend less time in and move through more quickly if the pace was determined by the user (Figure 6). These findings reiterate themes that are prevalent in existing literature, including the idea that varying levels of complexity influence the perceived duration of time (Shakibamanesh & Ghorbanian, 2017). Repetition, associated with the formality of the highly coherent scenarios, was linked to the linearity and direction of a path, making the path feel more direct and like a space meant to pass through, propelling the user forward. Similar results were found in another study (Kuper, 2017) where it was suggested that regular equidistant plantings do not encourage environmental exploration as much as clustered plants that may limit the visibility of environmental information. Connections found between components of the landscape and user experience emphasise the power of landscape architects and other outdoor space designers to create spaces to impact experience, reiterating the concept that the most powerful experiences occur when space, motion, orientation, and meaning reinforce each other.

Overarchingly, this study aligns with literature regarding time and landscapes in acknowledging a relationship between landscape elements and the perception of time, implying that designers possess the ability to affect user experience at the level of how a person feels that time passes and affording designers power over place experiences in this capacity (Bosselmann, 1998). Each design scenario may help achieve a unique design goal (Figure 7). If the objective of a design is to generate a satisfying and relaxing environment such as a higher education campus path through a quad, increased amounts of plant species and less structured organisation found in a high complexity and low coherence landscape might promote these program goals. High coherence paired with high complexity might be more favourable when creating a more formal space that still offers interest and depth to explore for those who choose to linger. To create a directed landscape that is not meant to engage the user but effectively move them to a specific point instead, slight variation in structure and plant species found in landscapes with high coherence and low complexity might be optimal. Other transitional spaces that are less directed but still lack engagement through interest might be achieved through a landscape that is low in both coherence and complexity, and lower visual interest with a relaxed layout may make a longer stroll along a secondary path more enjoyable.

Limitations and future work

There are numerous limitations in this research. First, experiential limitations exist in mimicking a real environment in virtual reality. Lack of audio and other sensory stimulus could inhibit

participants' ability to feel fully immersed in the VR environment. Another limitation lies with the novelty factor of virtual reality itself (George et al., 2022). Participants were shown the video twice to mitigate the newness of using VR equipment to most study participants. Furthermore, the experience of being moved through a space varies greatly from walking through the space oneself, even in VR. Future studies might re-assess the experiment in the physical built environment to confirm research outcomes.

Because of the qualitative focus of the study, there were fewer participants limiting the generalisation of findings but also strengthening the ability to have deep follow-up conversations. The findings could only be corroborated with existing literature, showing that VR can be a useful tool in studying people-place relations to better understand the effects of design choices. Future studies should incorporate more study participants to lead to more generalisable findings. Another limitation of this study is its restriction to faculty and students at the institution where the study took place, since landscape aesthetic preferences have been shown to vary between different groups of people (López-Rodríguez et al., 2019). Future studies should incorporate a broader audience to determine if results would vary. Additionally, qualitative coding of focus group discussions may have been influenced by the researchers' values and biases, and the researcher leading focus group discussions may have misinterpreted the tone and meaning of participants' dialogues.

Considering time and resources for this exploratory study, the 3D model used as the basis for this research was an existing model, and some participants had previous associations and interactions with the portion of the campus replicated. Some participants compared the model and experience to present real-life conditions of the site, which could have influenced how they perceived the path and the adjacent plantings. However, they also verified the realistic nature of the experience in VR. Future studies might develop environments fully unfamiliar to all study participants as a part of the experimental research design.

While efforts were made for gender diversity among study participants in each group, future research might delve further into the embodied experiences of study participants that may influence research outcomes, such as cultures, abilities, and backgrounds. Researchers might also investigate the meaning behind certain landscape perceptions, including an understanding of varying and evolving landscape design approaches. For instance, the study does not address how desirable landscape aesthetics might be used coincidentally with other aspects of landscape architecture, such as ecological landscape performance, to help shape people's preferences (Meyer, 2008). Rather, it tests human perceptions through the lens of existing landscape preference theories. Finally, this study incorporates four combinations of varying coherence and complexity to measure their effect on perceptions of time. However, it does not identify the particular point where these variables affect people's perception of time slowing down or speeding up. Future research might identify this point of change for more directed recommendations on design strategies.

Numerous studies have proposed ways to measure the qualities of landscape aesthetics (Jahani, 2019; Khalilnezhad et al., 2021; Ode et al., 2008; van der Jagt et al., 2014) yet emerging technologies such as VR provide new opportunities to reassess existing design theories. For example, while this study focuses on the landscape aesthetic facets of coherence and complexity, other aspects of landscape aesthetic theory such as preferences for naturalness in planting designs have not been tested using VR. Future research could try alternate methods in VR to see if the results are different. Participants could be allowed to navigate at an individual pace in VR to determine if the different environments would cause participants to spend more, less, or equal amounts of time in each scenario. Furthermore, additional research methods such as the collection of physiological data or surveys could be incorporated for a more quantitative approach.

New findings, including those implied in this study, could alter designers' approaches to creating places that invoke particular feelings and actions. As new knowledge emerges in

landscape architecture concerning the influence of landscape aesthetics, landscape architects and other outdoor space designers may intentionally enhance the application of this knowledge for the betterment of people and the environment. Additionally, they may explore novel approaches to integrate landscape aesthetics with other aspects of the field, redefining the meaning and consequence of aesthetic experiences. Despite the limitations, this study is one of the first of its kind to explore facets of landscape aesthetic theory using an immersive virtual environment, adding to the literature on designing landscapes with user experiences in mind and establishing the context for future research in this area.

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Informed consent

Informed consent was obtained from all research participants, and the study was approved by the University of Georgia Institutional Review Board.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

No data is available for this study, as qualitative data based on human subjects was stated to be made available only in an aggregated form and with select direct quotes.

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